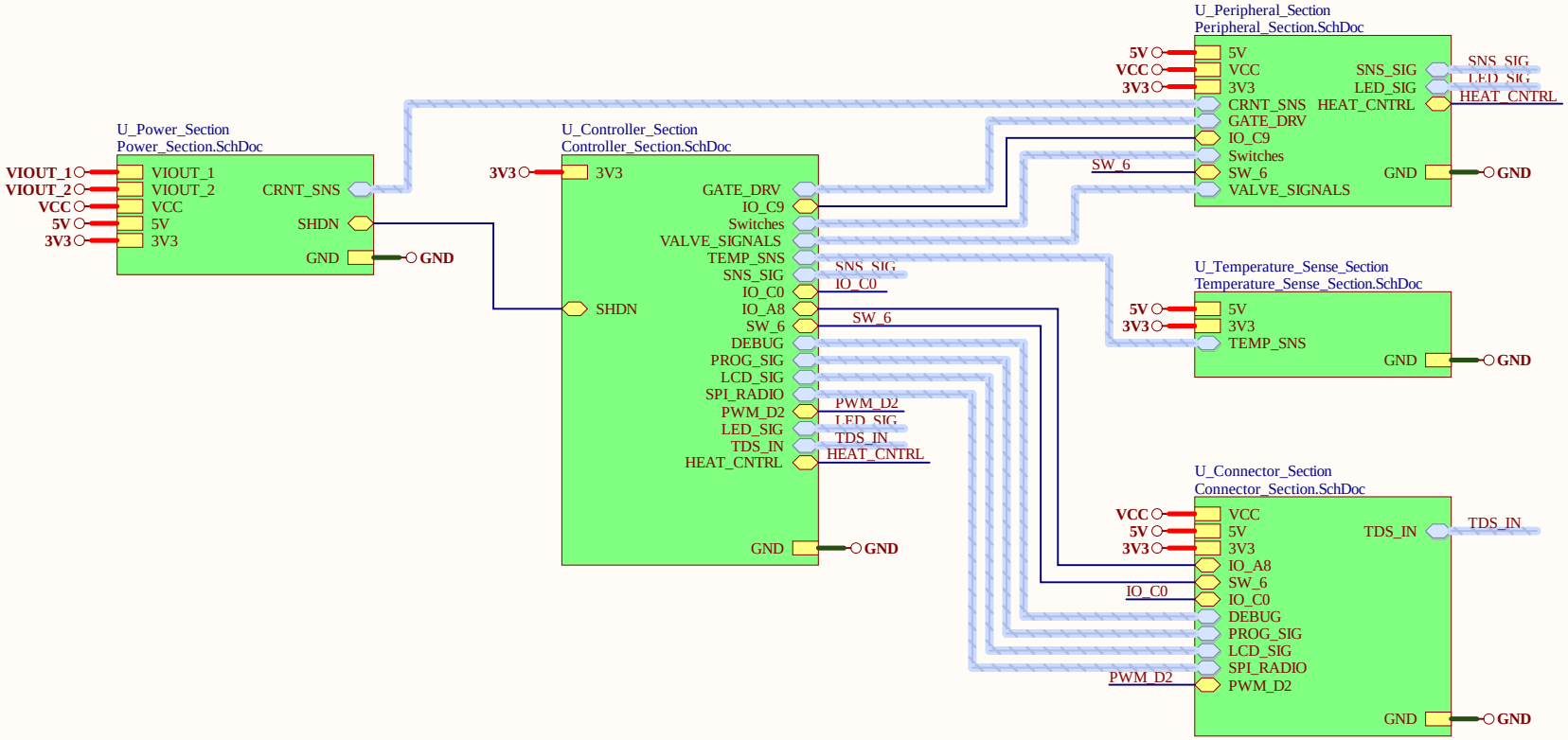
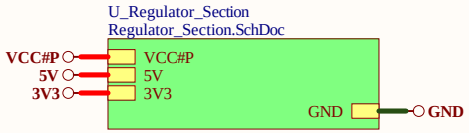
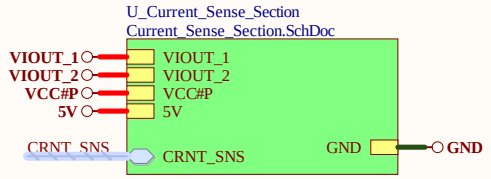
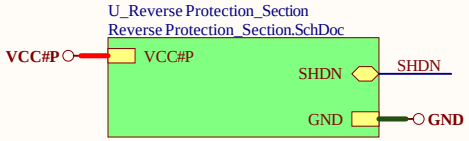
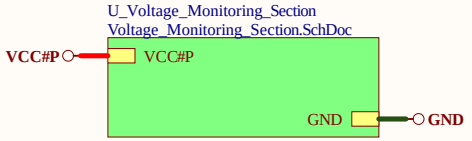

U_Power_Section
Power_Section.SchDoc
VIOUT_1
VIOUT_2
VCC
5V
3V3
VIOUT_1
VIOUT_2
VCC
5V
3V3
CRNT_SNS
SHDN
GND
GND
U_Controller_Section
Controller_Section.SchDoc
3V3
3V3
SHDN
GATE_DRV
IO_C9
Switches
VALVE_SIGNALS
TEMP_SNS
SNS_SIG
IO_C0
IO_A8
SW_6
DEBUG
PROG_SIG
LCD_SIG
SPI_RADIO
PWM_D2
LED_SIG
TDS_IN
HEAT_CNTRL
GND
GND
U_Peripheral_Section
Peripheral_Section.SchDoc
5V
VCC
3V3
5V
VCC
3V3
CRNT_SNS
HEAT_CNTRL
GATE_DRV
IO_C9
Switches
SW_6
VALVE_SIGNALS
SNS_SIG
LED_SIG
HEAT_CNTRL
GND
GND
SW_6
U_Temperature_Sense_Section
Temperature_Sense_Section.SchDoc
5V
3V3
5V
3V3
TEMP_SNS
GND
GND
U_Connector_Section
Connector_Section.SchDoc
VCC
5V
3V3
VCC
5V
3V3
IO_A8
SW_6
IO_C0
DEBUG
PROG_SIG
LCD_SIG
SPI_RADIO
PWM_D2
TDS_IN
GND
GND
IO_C0
PWM_D2
TDS_IN

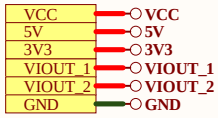

CRNT_SNS
CRNT_SNS
SHDN
SHDN


U_Regulator_Section
Regulator_Section.SchDoc
VCC#P
5V
3V3
VCC#P
5V
3V3
GND
GND


U_Current_Sense_Section
Current_Sense_Section.SchDoc
VIOUT_1
VIOUT_2
VCC#P
5V
CRNT_SNS
VIOUT_1
VIOUT_2
VCC#P
5V
CRNT_SNS
GND
GND

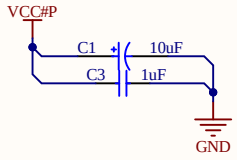

U_Reverse Protection_Section
Reverse Protection_Section.SchDoc
VCC#P
VCC#P
SHDN
SHDN
GND
GND


U_Voltage_Monitoring_Section
Voltage_Monitoring_Section.SchDoc
VCC#P
VCC#P
GND
GND

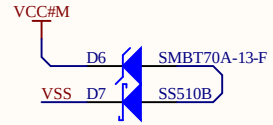

VCC
5V
3V3
VIOUT_1
VIOUT_2
GND
VCC
5V
3V3
VIOUT_1
VIOUT_2
GND

SHDN SHDN

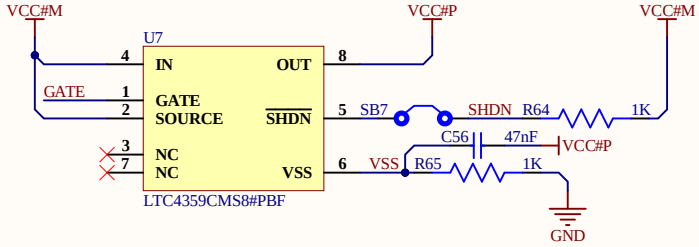
DECOUPLING CAPCITOR



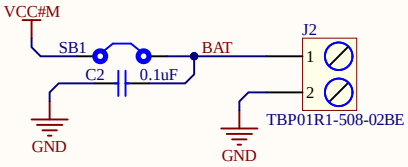
INPUT DIODES



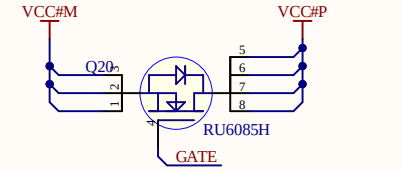
REVERSE POLARITY PROTECTION IC



BATTERY / MAIN INPUT

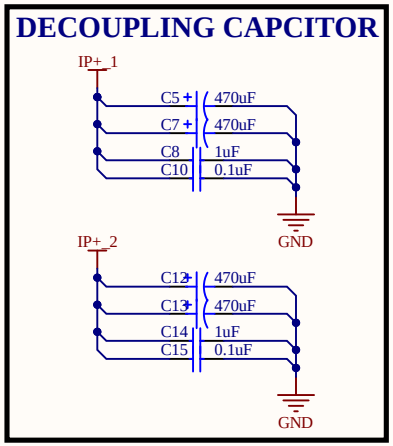


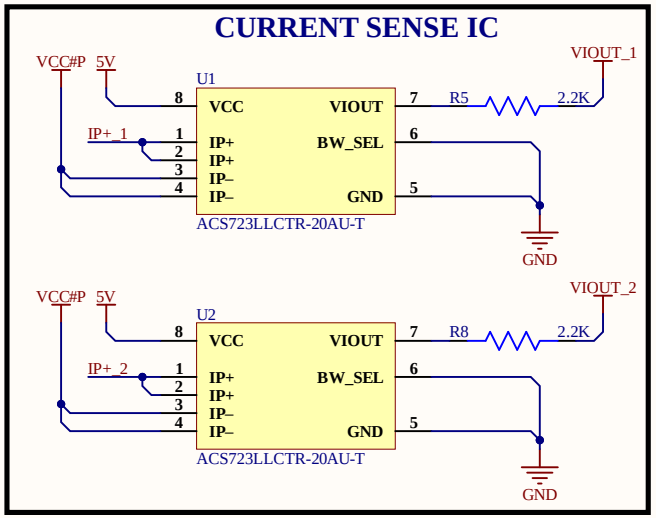
MOSFET

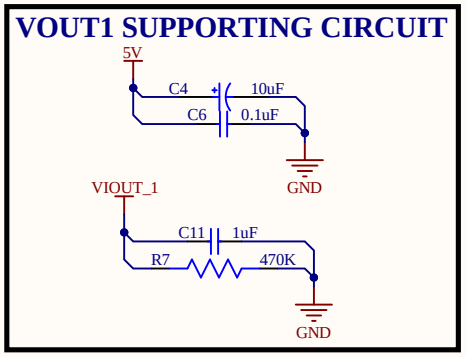


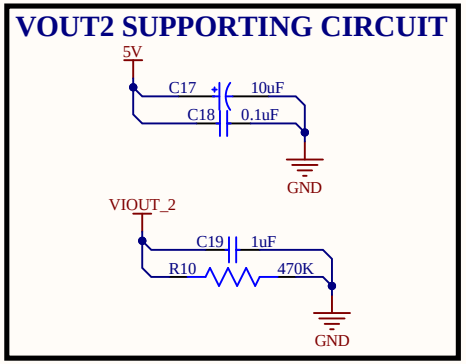
VCC#P
GND
VCC#P
GND

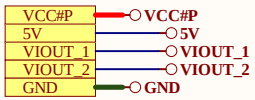

CRNT_SNS
CRNT_SNS
IP+_1
IP+_2
IP+_1
IP+_2


DECOUPLING CAPCITOR
IP+_1
C5
470uF
C7
470uF
C8
1uF
C10
0.1uF
GND
IP+_2
C12
470uF
C13
470uF
C14
1uF
C15
0.1uF
GND

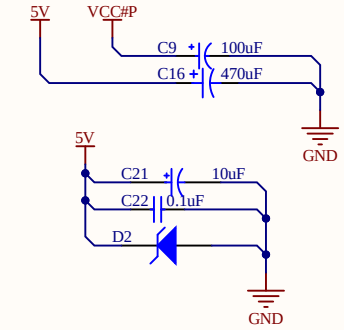

CURRENT SENSE IC
VCC#P
5V
U1
VCC
VIOUT
IP+
IP+
IP–
IP–
BW_SEL
GND
R5
2.2K
VIOUT_1
IP+_1
ACS723LLCTR-20AU-T
GND
VCC#P
5V
U2
VCC
VIOUT
IP+
IP+
IP–
IP–
BW_SEL
GND
R8
2.2K
VIOUT_2
IP+_2
ACS723LLCTR-20AU-T
GND

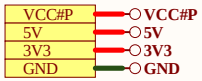

VOUT1 SUPPORTING CIRCUIT
5V
C4
10uF
C6
0.1uF
GND
VIOUT_1
C11
1uF
R7
470K
GND


VOUT2 SUPPORTING CIRCUIT
5V
C17
10uF
C18
0.1uF
GND
VIOUT_2
C19
1uF
R10
470K
GND

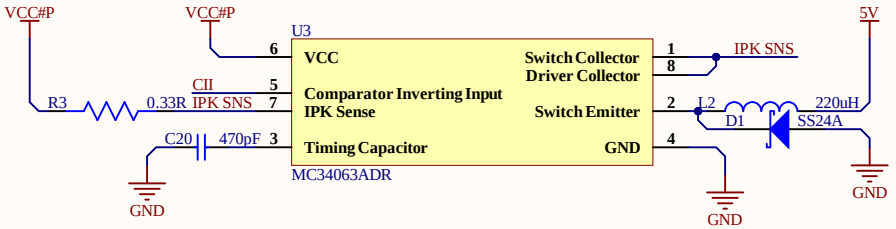

VCC#P
5V
VIOUT_1
VIOUT_2
GND
VCC#P
5V
VIOUT_1
VIOUT_2
GND

DECOUPLING CAPCITOR

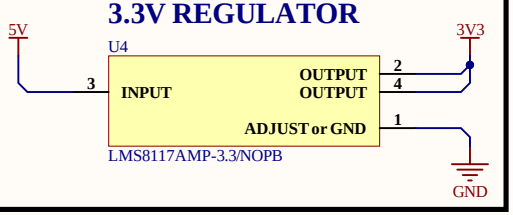




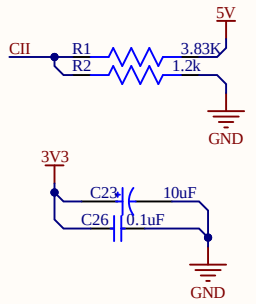
5V BUCK REGULATOR



3.3V REGULATOR

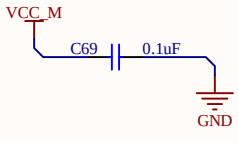


SUPPORTING CIRCUIT



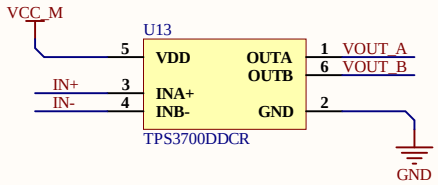


DECOUPLING CAPCITOR

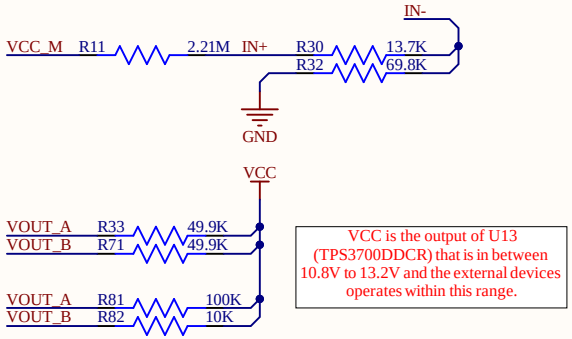




UNDER AND OVER VOLTAGE PROTECTION IC



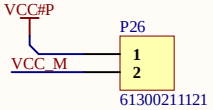
SUPPORTING CIRCUITRY



VCC is the output of U13 (TPS3700DDCR) that is in between 10.8V to 13.2V and the external devices operates within this range.

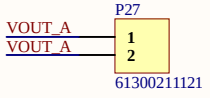
VOLTAGE TO MONITOR

Monitors Main 12V Input. Need shorting jumper to provide power to U13.



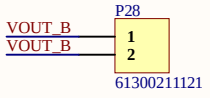
UNDER VOLTAGE OUTPUT

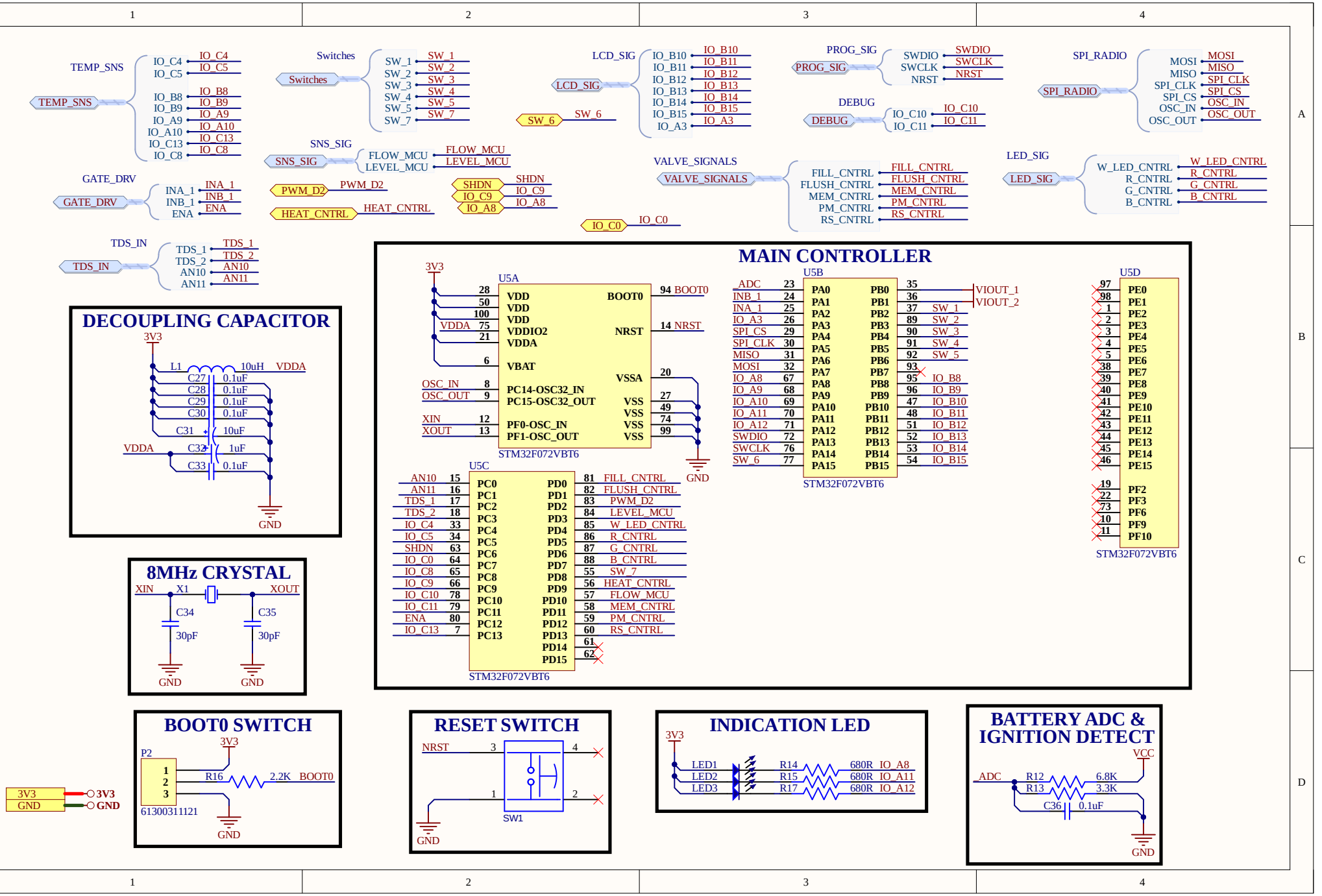
Under Voltage Open Drain Output. 100K Pullup is recommended.

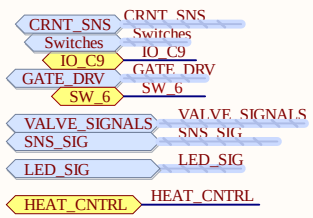


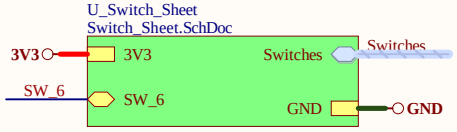
OVER VOLTAGE OUTPUT

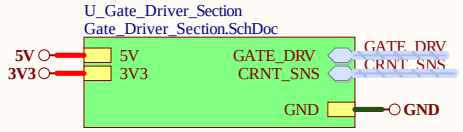
Over Voltage Open Drain Output. 10K Pullup is recommended.

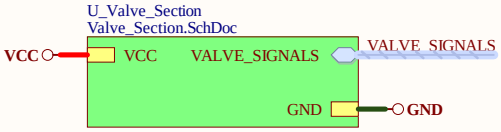


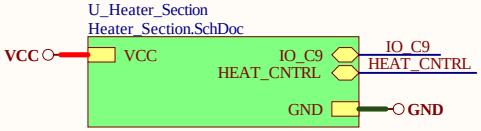

TEMP_SNS
IO_C4
IO_C5
IO_B8
IO_B9
IO_A9
IO_A10
IO_C13
IO_C8
Switches
SW_1
SW_2
SW_3
SW_4
SW_5
SW_7
LCD_SIG
IO_B10
IO_B11
IO_B12
IO_B13
IO_B14
IO_B15
IO_A3
SW_6
PROG_SIG
SWDIO
SWCLK
NRST
DEBUG
IO_C10
IO_C11
SPI_RADIO
MOSI
MISO
SPI_CLK
SPI_CS
OSC_IN
OSC_OUT
SNS_SIG
FLOW_MCU
LEVEL_MCU
GATE_DRV
INA_1
INB_1
ENA
PWM_D2
HEAT_CNTRL
SHDN
IO_C9
IO_A8
IO_C0
VALVE_SIGNALS
FILL_CNTRL
FLUSH_CNTRL
MEM_CNTRL
PM_CNTRL
RS_CNTRL
LED_SIG
W_LED_CNTRL
R_CNTRL
G_CNTRL
B_CNTRL
TDS_IN
TDS_1
TDS_2
AN10
AN11
DECOUPLING CAPACITOR
3V3
L1
10uH
VDDA
C27
C28
C29
C30
C31
C32
C33
0.1uF
10uF
1uF
GND
8MHz CRYSTAL
XIN
X1
XOUT
C34
C35
30pF
MAIN CONTROLLER
U5A
VDD
VDDIO2
VDDA
VBAT
PC14-OSC32_IN
PC15-OSC32_OUT
PF0-OSC_IN
PF1-OSC_OUT
BOOT0
NRST
VSSA
VSS
STM32F072VBT6
U5B
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
VIOUT_1
VIOUT_2
U5C
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PD8
PD9
PD10
PD11
PD12
PD13
PD14
PD15
U5D
PE0
PE1
PE2
PE3
PE4
PE5
PE6
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
PF2
PF3
PF6
PF9
PF10
BOOT0 SWITCH
P2
R16
2.2K
61300311121
RESET SWITCH
NRST
SW1
INDICATION LED
LED1
LED2
LED3
R14
R15
R17
680R
BATTERY ADC & IGNITION DETECT
VCC
ADC
R12
R13
6.8K
3.3K
C36
0.1uF

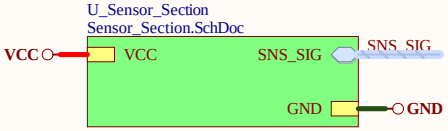

CRNT_SNS
Switches
IO_C9
GATE_DRV
SW_6
VALVE_SIGNALS
SNS_SIG
LED_SIG
HEAT_CNTRL

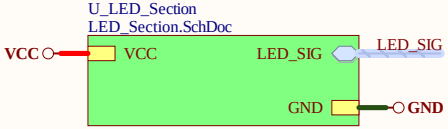

U_Switch_Sheet
Switch_Sheet.SchDoc
3V3
SW_6
3V3
SW_6
Switches
GND
Switches
GND

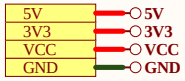

U_Gate_Driver_Section
Gate_Driver_Section.SchDoc
5V
3V3
5V
3V3
GATE_DRV
CRNT_SNS
GND
GATE_DRV
CRNT_SNS
GND

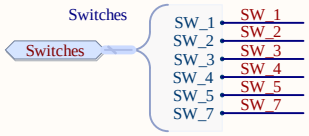

U_Valve_Section
Valve_Section.SchDoc
VCC
VCC
VALVE_SIGNALS
GND
VALVE_SIGNALS
GND


U_Heater_Section
Heater_Section.SchDoc
VCC
VCC
IO_C9
HEAT_CNTRL
GND
IO_C9
HEAT_CNTRL
GND

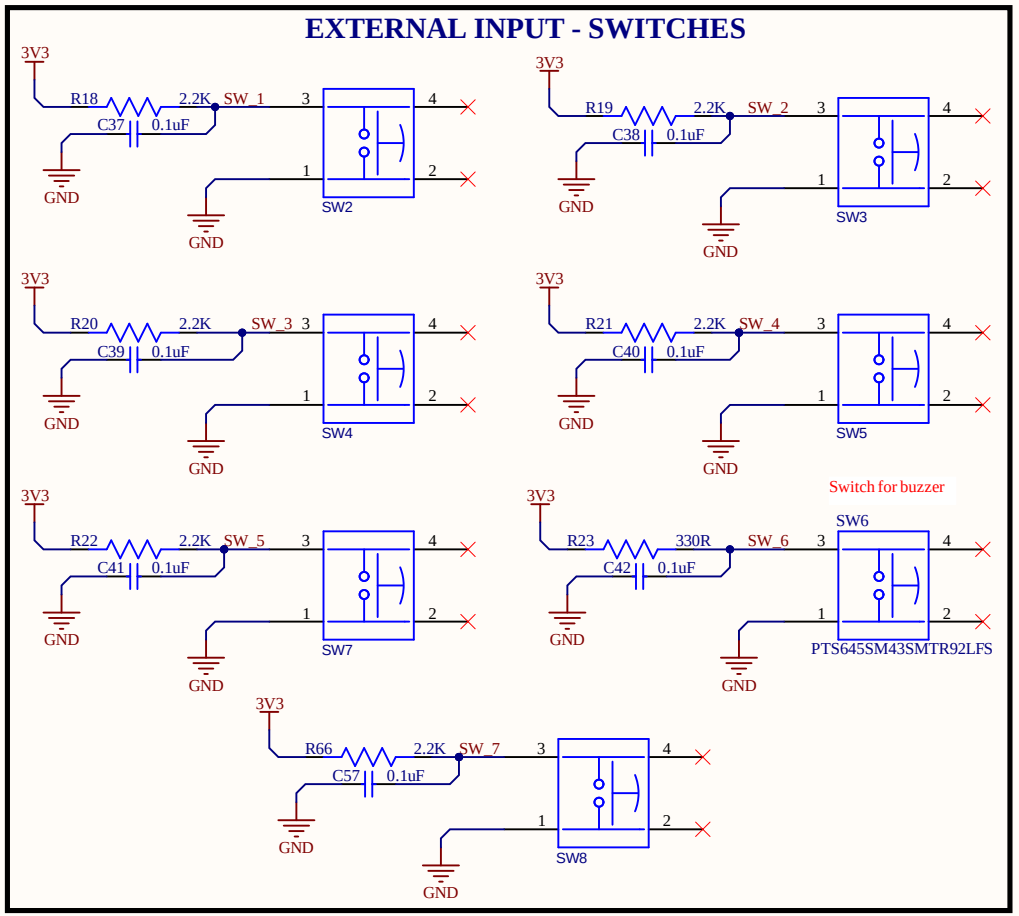

U_Sensor_Section
Sensor_Section.SchDoc
VCC
VCC
SNS_SIG
GND
SNS_SIG
GND

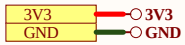

U_LED_Section
LED_Section.SchDoc
VCC
VCC
LED_SIG
GND
LED_SIG
GND

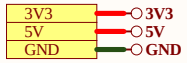

5V
3V3
VCC
GND
5V
3V3
VCC
GND

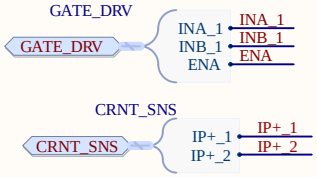

Switches
Switches
SW_1
SW_2
SW_3
SW_4
SW_5
SW_7
SW_1
SW_2
SW_3
SW_4
SW_5
SW_7


SW_6
SW_6

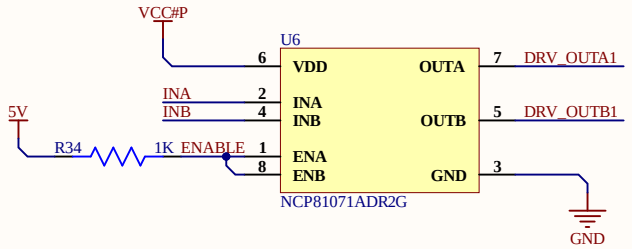

EXTERNAL INPUT - SWITCHES
3V3
R18
2.2K
SW_1
C37
0.1uF
GND
SW2
R19
SW_2
C38
SW3
R20
SW_3
C39
SW4
R21
SW_4
C40
SW5
R22
SW_5
C41
SW7
Switch for buzzer
SW6
R23
330R
SW_6
C42
PTS645SM43SMTR92LFS
R66
SW_7
C57
SW8


3V3
GND
3V3
GND

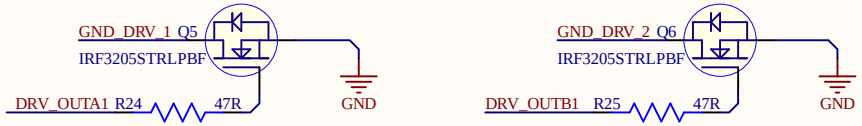




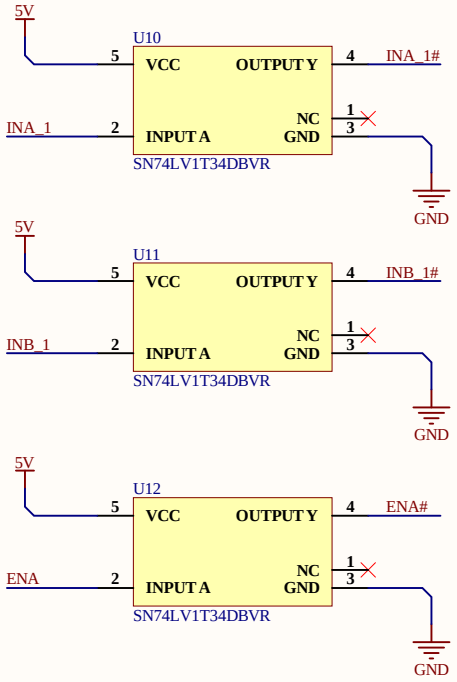
LOW SIDE PUMP DRIVER IC



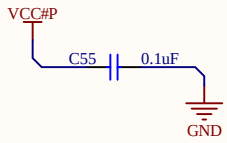
MOSFET DRIVERS



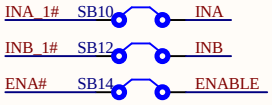
LEVEL SHIFTER



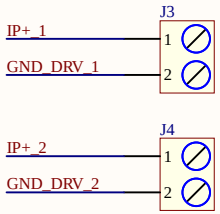
DECOUPLING CAPACITOR



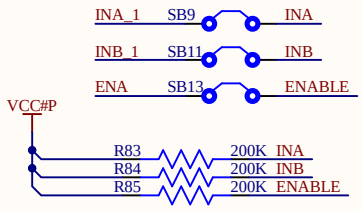
USING LEVEL TRANSLATR



HEADERS FOR PUMP

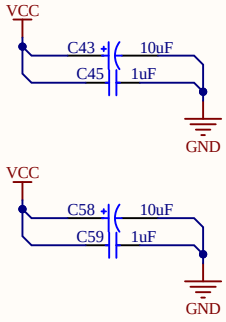


W/O USING LEVEL TRANSLATR

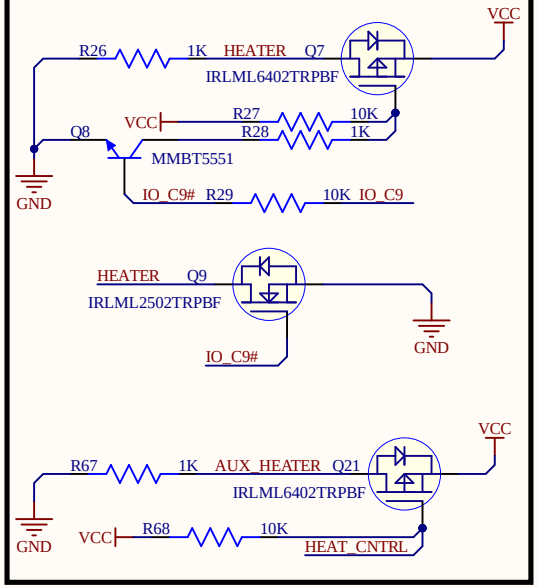




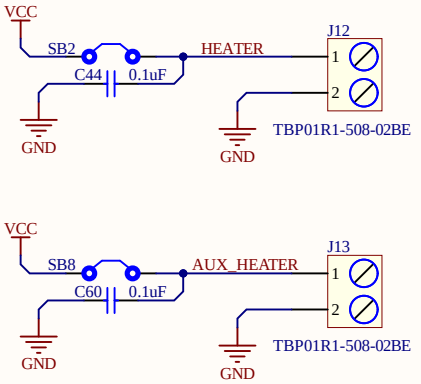
DECOUPLING CAPACITOR



MOSFET CIRCUITRY



HEATER INPUT





VALVE_SIGNALS

VALVE_SIGNALS

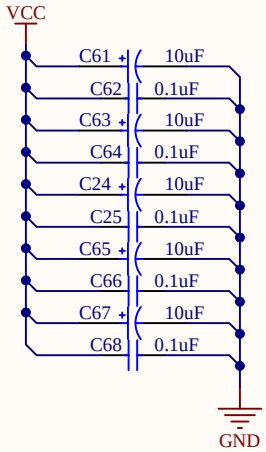
FILL_CNTRL
FLUSH_CNTRL
MEM_CNTRL
PM_CNTRL
RS_CNTRL

FILL_CNTRL
FLUSH_CNTRL
MEM_CNTRL
PM_CNTRL
RS_CNTRL

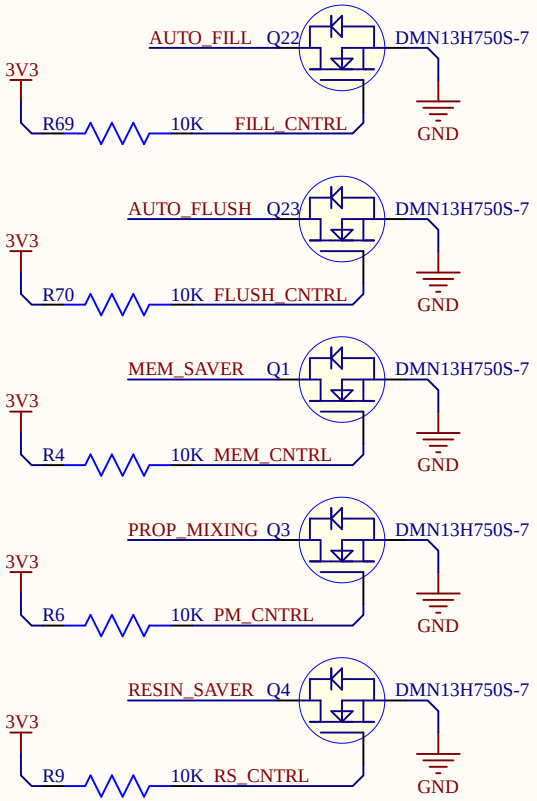
VCC
GND

VCC
GND

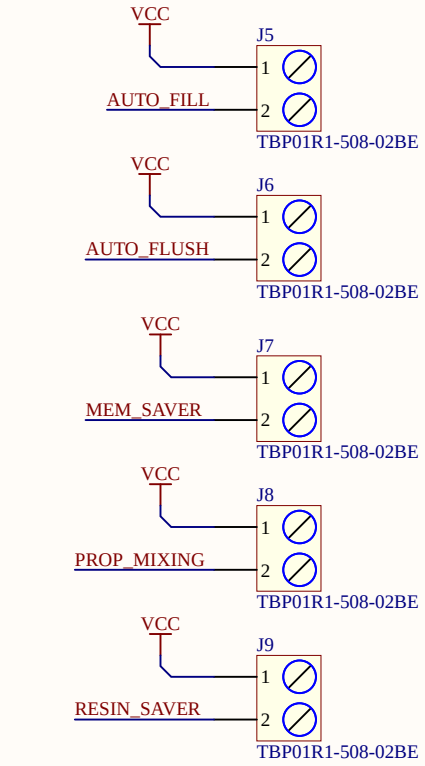
DECOUPLING CAPACITOR



SOLENOID CONTROL CIRCUIT



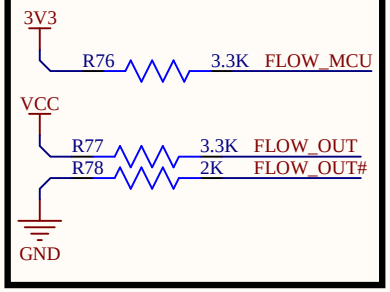
SOLENOID VALVE INPUTS



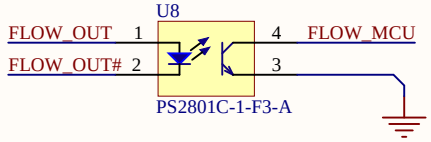
Title		
Size A	Number	Revision
Date: 7/24/2022		Sheet of
File: C:\Users\..\Valve_Section.SchDoc		Drawn By:

SNS_SIG
SNS_SIG
FLOW_MCU
LEVEL_MCU
FLOW_MCU
LEVEL_MCU

PULL UP RESISTORS

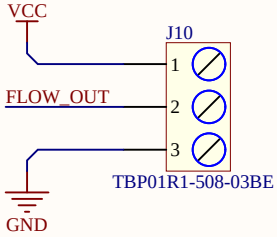


OPTO ISOLATION FOR FLOW METER

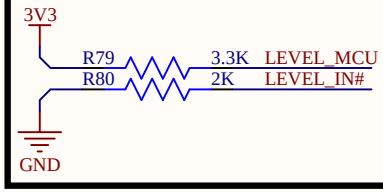


Forward voltage of diode=1.2V
With VCC=12V, IF=5.4mA
Flow meter datasheet recommends a pull up value between 3K-30K

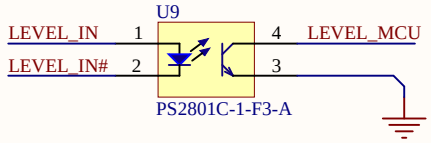
FLOW METER INPUT



SUPPORTING CIRCUIT

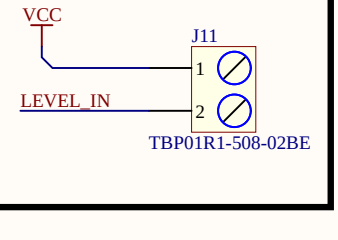


OPTO ISOLATION FOR FLOW METER



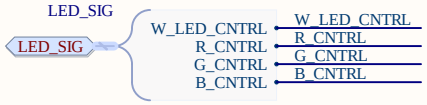
Forward voltage of diode=1.2V
With VCC=12V, IF=54mA

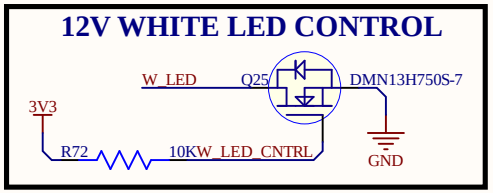
LEVEL SENSOR INPUT

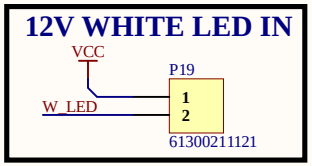


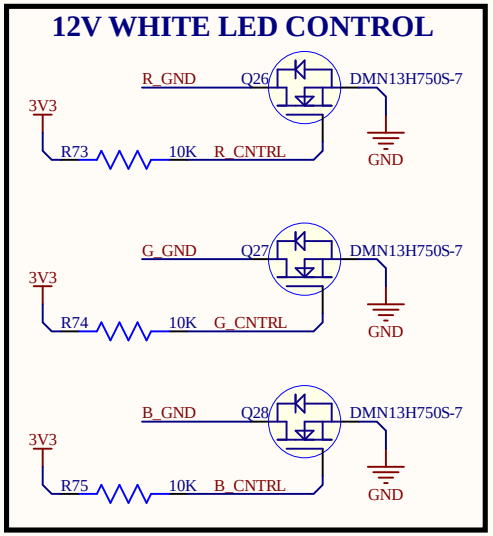
VCC
GND
VCC
GND

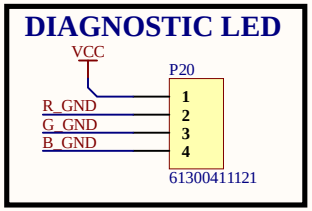
Title		
Size: A	Number	Revision
Date: 7/24/2022		Sheet of
File: C:\Users\..\Sensor_Section.SchDoc		Drawn By:







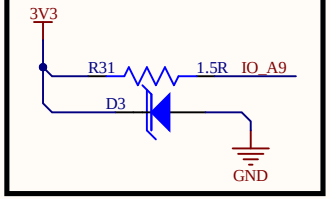




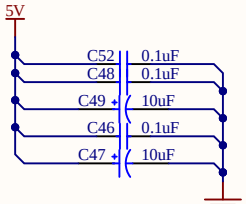


Title		
Size: A4	Number	Revision
Date: 7/24/2022		Sheet of
File: C:\Users\..\LED_Section.SchDoc		Drawn By:

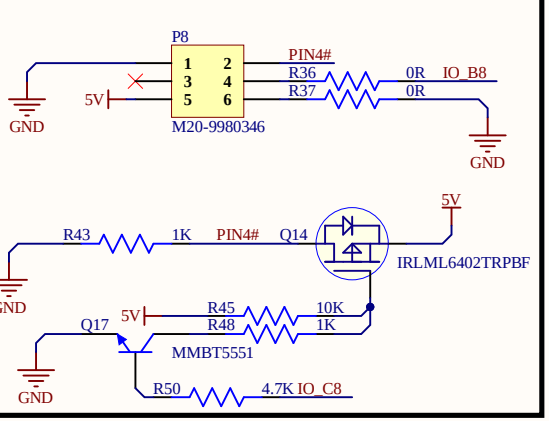
PULL UP RESISTORS

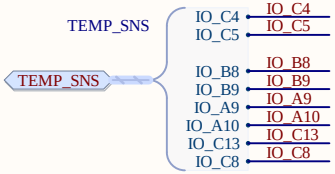


DECOUPLING CAPACITOR

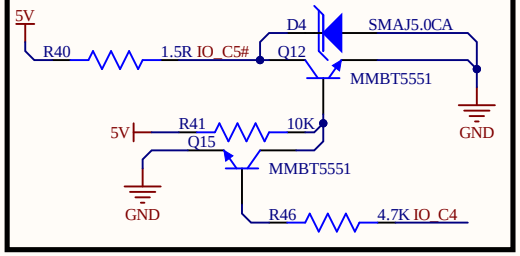


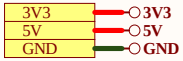
1-WIRE TEMPERATURE SENSOR



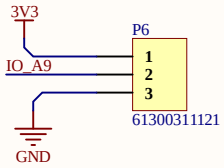


SUPPORTING CIRCUITRY

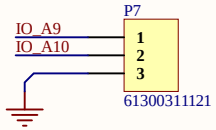




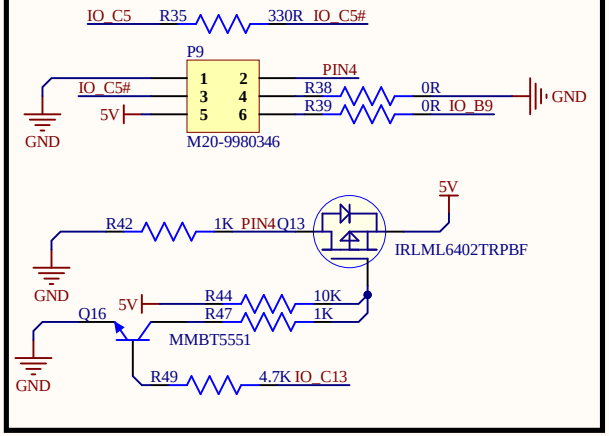
TEMPERATURE SENSOR DS18B20

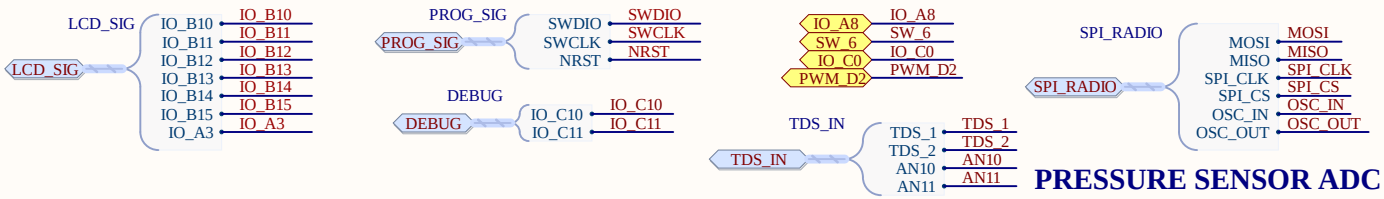


BOOT LOADER



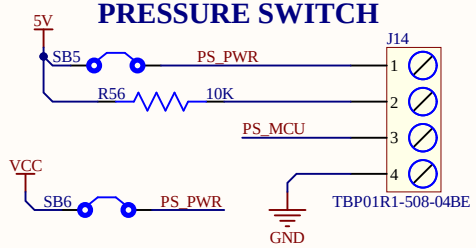
1-WIRE TEMPERATURE SENSOR

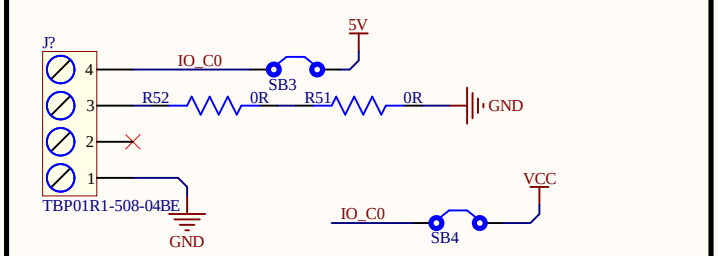




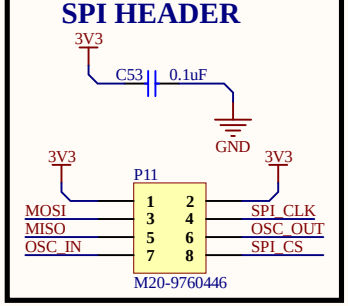
PRESSURE SENSOR ADC

PRESSURE SWITCH

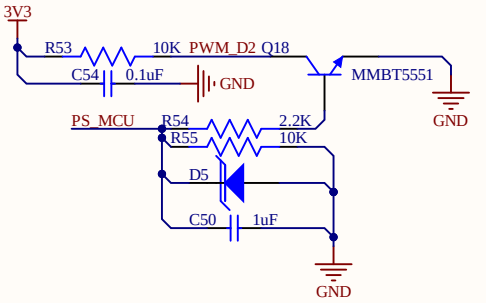




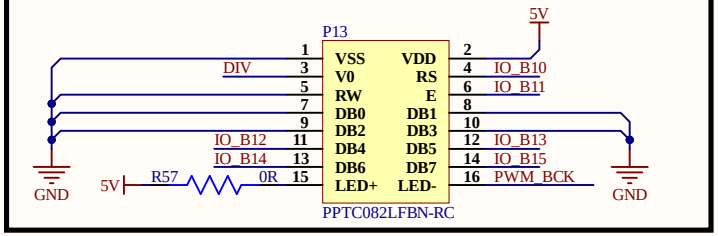
SPI HEADER



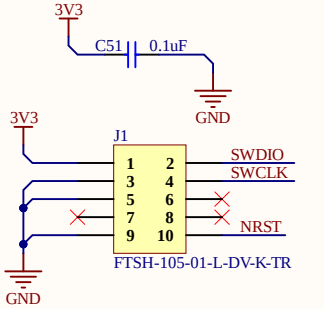
SUPPORTING CIRCUIT



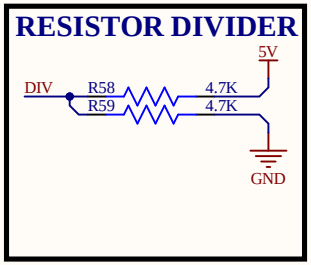
LCD CONNECTOR



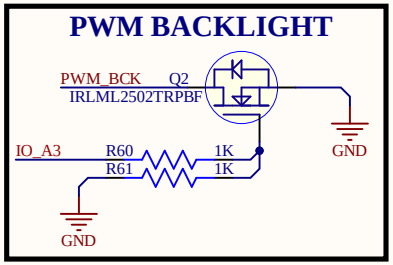
SW PROGRAMMING



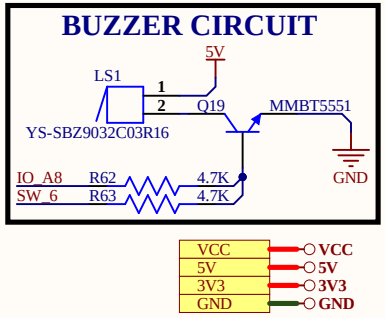
RESISTOR DIVIDER



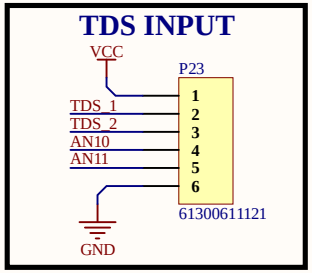
PWM BACKLIGHT



BUZZER CIRCUIT



TDS INPUT



FTDI 3V3 DEBUGGING HEADER

